Siddhant A. Deshmukh, PhD

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PROFESSIONAL EXPERIENCE

Master Data Manager - *SBM Life Science Corp (Cary, NC)* 05/2024 - present

- Managed and governed master data for the entire organization to ensure accuracy and consistency across departments
- Led effort to write a comprehensive Master Data Governance policy for the entire US organization
- Developed autonomous processes with Supply Chain, Sales & Customer Service to reduce errors on incoming orders by over 80%, significantly improving the order-to-cash flow
- Spearheaded collaboration with Marketing to develop product launch and product transition processes, achieving over 95% on-time project delivery
- Partnered with Sales to establish pricing-related processes, reducing pricing errors by 90%
- Managed and mentored an employee, fostering their autonomy and professional growth while driving improvements in retailer syndication

Co-Founder & Lead Data Scientist - *sci-an (Max Planck Institute for Astronomy, Heidelberg, Germany)* 01/2022 - 10/2023

- Secured over €400,000 in funding as part of a four-member team for an entrepreneurial venture focused on analyzing customer feedback at large conferences and trade fairs across Europe
- Delivered comprehensive, data-driven reports for clients, leveraging LLMs to generate key takeaways, which boosted customer retention by 70%
- Conducted advanced text data analysis using dimensionality reduction, NLP techniques, and clustering algorithms (UMAP, MDE, scipy, scikit-learn) to uncover actionable insights
- Designed and implemented key components of an online virtual networking tool using React.js, enhancing user engagement
- Created interactive dashboards using R, Python (Dash), and React to visualize insights and improve decision-making for clients
- Led social media marketing and targeted sales campaigns, driving a 50% increase in customer acquisition

Astrophysics Researcher - *Heidelberg University (Heidelberg, Germany)* 08/2019 - 12/2022

- Enhanced spectroscopic fitting routines and collaborated internationally to develop a rigorous analysis pipeline for determining the solar photospheric silicon abundance using Python
- Developed a codebase for coupling chemical reaction systems with hydrodynamics and radiative transfer models using Python, Julia, and Fortran
- Created novel analysis tools leveraging pathfinding algorithms on weighted, directed graphs to identify reaction pathways in chemical networks with networkx
- Developed & trained deep learning models to predict equilibrium chemistry (Pytorch, TensorFlow, Keras) resulting in a 20x speed-up of computations
- Led two independent research projects in computational astrophysics, culminating in publications in esteemed, peer-reviewed journals

EDUCATION

PhD in Astrophysics - *Heidelberg University (Heidelberg, Germany)* 2019 - 2023

- Thesis: "Modelling Non-Equilibrium Molecular Formation and Dissociation for the Spectroscopic Analysis of Cool Stellar Atmospheres"
- Computational modelling and data analysis for coupling chemical reaction systems to hydrodynamics and radiative transfer models using Fortran, Python, C, Julia

Integrated Master's Degree in Physics (First-Class Honours) - *University of Exeter (Exeter, UK)* 2015 - 2019

- Thesis: "Refining Computational Models of Stellar Rotation Evolution Using Open Magnetic Flux"
- Analysis of stellar magnetic field evolution by solving differential equations in C with data visualization in Python

KEY SKILLS

Programming Languages: Python (expert); C, JavaScript, Julia, R, Rust, SQL (intermediate); Java (basic)
Software: Linux, Git, GitHub, Microsoft Office, PowerBI, LaTeX, vim
Languages: English (native); German, Hindi, Marathi (working proficiency)

LEADERSHIP EXPERIENCE

Master Data Manager - *SBM Life Science Corp (Cary, NC)* 05/2024 - present

- Supervised & mentored an employee, fostering their autonomy & professional growth, resulting in improved retailer syndication and process efficiency
- Led cross-functional collaborations with Marketing, Sales, and Supply Chain teams to establish data-driven processes, reducing errors by 80% and fines by 30%

Research Supervisor - *DAAD RISE Germany (Heidelberg, Germany)* 06/2021 - 09/2021

- Earned funding for 3 months of supervising an American undergraduate exchange student in the German research environment

PhD Tutor - *Heidelberg University (Heidelberg, Germany)* 2019 - 2023

- Created material and ran tutorials for undergraduate and master's level courses in English and German including "Physics for Non-Physicists" and "Fundamentals of Simulation Methods"

Conference Presentations

- Cambridge Workshop of Cool Stars, Stellar Systems and the Sun. Toulouse, France (July 2022)
- Planetary Transits and Oscillations Stellar Science Conference. Barcelona, Spain (Sept 2019)

Attended Schools and Workshops

- 47th Heidelberg Physics Graduate Days. Heidelberg, Germany (Oct 2021)
- International Max Planck Research Schools' Summer School on Stellar Ecosystems. Heidelberg, Germany (Sept 2021)
- KROME Summer School Workshop on Astrochemistry. Online (Feb 2021)

PUBLICATIONS

- **Siddhant A. Deshmukh**, Hans-Guenter Ludwig and Guillaume Guiglion (in prep). *Time-Dependent Molecular Chemistry in Red Giant Stellar Atmospheres with Neural Networks.*
- **Siddhant A. Deshmukh** (Oct 2023). *Modelling Non-Equilibrium Molecular Formation and Dissociation for the Spectroscopic Analysis of Cool Stellar Atmospheres*. PhD Thesis.
- **Siddhant A. Deshmukh** and Hans-Guenter Ludwig (July 2023). *Implications of Time-Dependent Chemistry in Metal-Poor Dwarf Stars.* Astronomy & Astrophysics Volume 675, A146.
- **Siddhant A. Deshmukh**, Hans-Guenter Ludwig, Arunas Kučinskas, Matthias Steffen, Paul S. Barklem, Elisabetta Caffau, Vidas Dobrovolskas, and Piercarlo Bonifacio (Dec 2022). *The Solar Photospheric Silicon Abundance According to CO5BOLD - Investigating Line Broadening, Magnetic Fields, and Model Effects.* Astronomy & Astrophysics Volume 668, A48.
- Adam J. Finley, **Siddhant A. Deshmukh**, Sean P. Matt, Mathew Owens, and ChiJu Wu. (Sept 2019). *Solar Angular Momentum Loss over the Past Several Millennia.* The Astrophysical Journal, Volume 883, Number 1.

ADDITIONAL INFORMATION

Interests: music, martial arts, game development, creative writing
Work Eligibility: U.S. citizen; eligible to work in the U.S. with no restrictions